D DigestAnything

RALPHTHON SEOUL #2

SOLO BUILD

DigestAnything

Any source → **See it. Test it. Prove it.**

Paste a URL, YouTube link, or PDF → get 5 AI-powered learning experiences

Builder
Jaeho Shin

Date
March 29, 2026

GitHub
jays0606/DigestAnything

PROBLEM

Passive Learning Doesn't Stick

80%

of content forgotten within 24 hours

— Ebbinghaus Forgetting Curve

Every day, millions read articles, watch videos, skim papers. **Feels productive. Almost nothing retained.**

The Real Pain

- ✕ No active recall — just highlighting and scrolling
- ✕ No self-testing — can't explain what you just read
- ✕ No reinforcement — content consumed once, forgotten forever

> *"I watched a 1-hour Karpathy lecture. Next day, I couldn't explain a single concept to a colleague."*

SOLUTION

5 Learning Experiences, 1 Click

Paste any URL, YouTube link, or PDF → get 5 AI-powered experiences grounded in **learning science**.

🧠 Overview

Summary + interactive mindmap

Dual Coding Theory

❓ Quiz

10 MCQs with explanations

Active Recall

🃏 Cards

12-15 flip flashcards

Spaced Repetition

🎙️ Podcast

2-host AI podcast with audio

Elaborative Interrogation

🎓 Tutor

Socratic Feynman tutor

Feynman Technique

💬 Chat

Ask anything, always available

On-demand Reinforcement

ARCHITECTURE

One Ingest, Parallel Everything

Pipeline

1 **Source Input**
URL / YouTube / PDF

↓

2 **Gemini 3 Flash Analysis**
→ context.json (concepts, hierarchy, insights)

↓

asyncio.gather() — ALL PARALLEL

Visual | Quiz | Cards | Podcast

Stack

Frontend	Next.js 14 + Tailwind
Backend	FastAPI (async)
AI	Gemini 3 Flash + TTS
Design	Google Stitch
Diagrams	Markmap
Agent	Claude Code + oh-my-claudecode

RALPH SETUP

Self-Healing Agent Loop

The secret sauce: **spec-first, test-first, then autonomous build with self-correcting agents.**

The Loop

1. **Write Spec + Tests First**
   895-line SPEC.md + 17 test files before any code

↓

2. **Autopilot Builds 4 Rounds**
   Claude Code /autopilot → BE + FE in parallel

↓

3. **Agent Test Loop (Opus 4.6)**
   Test → Fail → Opus reads error → Fixes → Retest

↓

4. **Playwright E2E + Audio QA**
   Browser test every tab + Gemini validates podcast audio

What Makes It "Ralph"

Spec-First Architecture

895 lines of spec, 504 lines of verified API reference, Stitch design system — all written BEFORE implementation

Claude Agent SDK Self-Correction

Opus 4.6 agents read test failures + source code → make targeted fixes → retest automatically. No human intervention.

4-Stage Verification

API tests → Agent fix loop → Playwright E2E → Gemini audio analysis. All automated, all in one run.

DEMO

Let's Digest Something

Try it with

📝 Anthropic Engineering Blog

anthropic.com/engineering/harness-design

🎬 Karpathy YouTube Talk

youtube.com/watch?v=kwSVtQ7dziU

~10s	5	100%
Overview loads	Learning tabs	Science-backed